# A MEASURABLE BICYCLE THEOREM

ANONYMOUS

Abstract. Let $T(u)$ and $T(v)$ be unit circles whose centres satisfy $1 < |u - v| < 2$. Voronov's bicycle theorem says that $T(u) \cup T(v)$ cannot be covered by three closed sets none of which realises distance 1. We prove a measurable analogue that needs no regularity of the colour classes. Let $\Omega$ be any open neighbourhood, however thin, of $T(u) \cup T(v)$, or more generally of two curves that are $C^1$-close to such circles. Then $\Omega$ has no *weak measurable* 3*-colouring*: there is no measurable $f\colon \Omega \to \{1, 2, 3\}$ whose monochromatic unit pairs form a null set of pairs. The proof combines four ingredients.

(i) A topological index for colourings that are locally 2-valued. This gives integer step counts with no perimeter assumption.
(ii) A "shell lemma" and a "cone lemma" for the set $\Sigma_3$ where all three colours accumulate. We use them to show that $\Sigma_3$ is 0-dimensional inside the band.
(iii) The removability of 0-dimensional singular sets for step counts.
(iv) A complete analytic description of the unit-distance correspondence between two unit circles, including the fold that occurs for $|u - v| > \sqrt{3}$, together with its $C^1$-stability.

The result is one of the steps of a Sokolov–Voronov-type argument for weak measurable 6-colourings of the plane. By OpenAI's transfer theorem, such an argument would decide the Hadwiger–Nelson problem. We stress that this paper proves only this one step. The remaining steps, which concern the supply of trichromatic points, are open.

## Contents

1. Introduction 1
1.1. Background 1
1.2. The measurable problem 2
1.3. Relation to the Hadwiger–Nelson problem 3
1.4. Strategy of the proof 3
2. Preliminaries 4
2.1. Notation 4
2.2. Weak colourings of open sets 4
2.3. Lebesgue points and essential supports 5
2.4. Two-colour rigidity 5
2.5. A thin-intersection estimate 6
3. The geometry of two unit circles 6
3.1. Exact circles 7
3.2. Stability of regular level sets 9
3.3. Near bicycles 10
3.4. Room in the band 11
4. Step counts off $\Sigma_3$ 11
4.1. The colour covering 11
4.2. The pair map 12
5. The shell lemma and the interior of $\Sigma_3$ 13

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5.1. Scale matching 13
5.2. The shell lemma 13
5.3. Empty interior 15
6. $F$-regions, cones, and the dimension of $\Sigma_3$ 15
6.1. The cone lemma 15
6.2. $\Sigma_3$ is zero-dimensional 16
7. Removability of $\Sigma_3$ 17
8. Proof of the Main Theorem 18
Concluding remarks 18
References 19

## 1. Introduction

1.1. **Background.** The chromatic number $\chi(\mathbb{R}^2)$ of the plane is the least $k$ for which $\mathbb{R}^2$ can be coloured with $k$ colours so that no two points at distance exactly 1 have the same colour.

*Classical bounds.* The Moser spindle and Isbell's hexagonal colouring give $4 \leqslant \chi(\mathbb{R}^2) \leqslant 7$. De Grey's 5-chromatic unit-distance graph [1] improved the lower bound to 5; see also [2].

*Measurable colourings.* For colourings with Lebesgue measurable classes, Falconer [3] proved that at least 5 colours are needed.

*OpenAI's results.* Recently OpenAI [5] proved $\chi(\mathbb{R}^2) \geqslant 6$ with no regularity assumption. Their argument has two parts.

- A *transfer theorem* [5, Thm. 1.3]. For every $k$, a proper $k$-colouring of $\mathbb{R}^2$ exists if and only if a *weak measurable $k$-colouring* exists. Such a colouring is a measurable map $c\colon \mathbb{R}^2 \to \{1,\dots,k\}$ with
$$\sum_i \int_{B(0,R)} \int_{S^1} \mathbf{1}_{A_i}(x)\mathbf{1}_{A_i}(x+u)\, d\sigma(u)\, dx = 0 \qquad \text{for all } R > 0.$$
- A geometric proof that no weak measurable 5-colouring exists.

So the Hadwiger–Nelson problem is now equivalent to a question about weak measurable colourings:

$$\chi(\mathbb{R}^2) = 7 \iff \text{there is no weak measurable 6-colouring of } \mathbb{R}^2.$$

*Regular colourings.* For colourings with regular colour classes, seven colours are known to be necessary in several settings:

- planar maps (Woodall [10], Townsend [7, 8]);
- Sokolov and Voronov [6], for map-type colourings under finiteness conditions;
- Voronov [9], for colourings in which each colour class avoids a whole interval $(1-\varepsilon, 1+\varepsilon)$ of distances. Equivalently, by compactness, closed covers of a disc of radius 3 by six closed sets cannot avoid distance 1.

*The bicycle step.* The proofs in [6, 9] share an index argument for *two* unit circles. In [9] it appears as the bicycle theorem: for $1 < |u-v| < 2$, the union $T(u) \cup T(v)$ admits no 3-colouring in which every colour class avoids all distances in some interval $(1-\varepsilon, 1+\varepsilon)$. In [6] its role is played by Lemma 3: two trichromatic points with the same multicolour are never at distance in $(1, 2)$.

The index argument runs as follows.

- Colour changes along a curve get signs in $\mathbb{Z}/3$.
- When two points move while staying at unit distance, the pair of colours walks on a 6-cycle. As a result, their index changes differ by at most 1.

- One moves a point of $T(u)$ to an intersection point $p \in T(u) \cap T(v)$ while tracking its two unit partners on $T(v)$. This forces an index of absolute value $\leqslant 1$ on a $120^\circ$ arc of $T(v)$.
- Chains of $60^\circ$ arcs force absolute value $\geqslant 2$ on the same arc.

1.2. **The measurable problem.** For weak measurable colourings, none of the regularity used in [6, 9] is available:

- colour classes need not be closed, and need not have finite perimeter;
- their topological boundaries may have positive area;
- unit circles may meet colour boundaries in Cantor sets;
- the hypothesis says nothing about any single circle, which is a null set of pairs.

In fact, colourings of thin annuli can contain arbitrarily fine radial "shells" of two colours, and fat Cantor-type interfaces.

The natural measurable replacement of the bicycle theorem is therefore a statement about two-dimensional *bands*. This paper proves it.

**Definition 1.1.** Let $1 < d < 2$ and $\kappa > 0$. A *$\kappa$-near bicycle* is a pair of curves

$$\Gamma_u = \{u + \rho_u(\theta)\mathbf{e}(\theta)\}, \qquad \Gamma_v = \{v + \rho_v(\theta)\mathbf{e}(\theta)\} \qquad (\theta \in \mathbb{R}/2\pi\mathbb{Z})$$

with the following properties:

- $|u - v| = d$, where $\mathbf{e}(\theta) = (\cos\theta, \sin\theta)$;
- $\rho_u, \rho_v \in C^1(\mathbb{R}/2\pi\mathbb{Z})$;
- $\|\rho_\bullet - 1\|_{C^1} := \sup|\rho_\bullet - 1| + \sup|\rho'_\bullet| \leqslant \kappa$.

A pair of curves that is $C^2$-close to two unit circles is in particular a $\kappa$-near bicycle.

**Main Theorem** (Theorem 8.1)**.** For every $d \in (1, 2)$ there is $\kappa_0(d) > 0$ with the following property. Let $(\Gamma_u, \Gamma_v)$ be a $\kappa$-near bicycle with $\kappa \leqslant \kappa_0$, let $\delta > 0$, and let $\Omega \supseteq N_\delta(\Gamma_u \cup \Gamma_v)$ be open. Then $\Omega$ admits no weak measurable 3-colouring.

Here $N_\delta$ denotes the open $\delta$-neighbourhood, and weak colourings of open sets are defined in Definition 2.1. Note that $\delta$ may be much smaller than $\kappa$.

**Corollary.** Let $1 < |u-v| < 2$ and $\delta > 0$. Then $N_\delta(T(u)) \cup N_\delta(T(v))$ admits no weak measurable 3-colouring.

**Remark 1.2** (What is new and what is not)**.**

- The theorem is a measurable version of the bicycle theorem of [9], and of the index lemma (Lemma 3) of [6].
- For colourings whose classes have locally finite perimeter, the theorem follows quite directly from the classical index argument. One applies BV slicing to almost every translate of the bicycle; see Remark 8.2.
- The point of this paper is to remove every regularity assumption. The main difficulty is the closed set $\Sigma_3$ of points near which all three colours have positive measure. We show three things about it:
  - off $\Sigma_3$ a colouring is locally 2-valued, and this already gives a topological index (Section 4);
  - inside the band, $\Sigma_3$ contains no continuum (Sections 5–6);
  - 0-dimensional singular sets are removable for the index (Section 7).

1.3. **Relation to the Hadwiger–Nelson problem.** We are explicit about the status of this result. It does *not* decide whether $\chi(\mathbb{R}^2)$ is 6 or 7.

Here is how it fits into a possible proof of $\chi(\mathbb{R}^2) = 7$. By the transfer theorem, it suffices to exclude weak measurable 6-colourings. A natural strategy is to transplant the Sokolov–Voronov argument [6] to the measurable setting. That argument has four steps:

(SV1) Around a trichromatic point $x$ with multicolour $T$, the three "outsider" colours $\{1, \dots, 6\} \setminus T$ fill a band around a closed curve near $\omega(x)$, the unit circle about $x$.
(SV2) Two trichromatic points with disjoint multicolours are at distance $\geqslant 2$.
(SV3) Two trichromatic points with the same multicolour are not at distance in $(1, 2)$.
(SV4) A combinatorial lemma in a disc of radius 3 produces a pair of trichromatic points that violates (SV2) or (SV3).

Suppose a measurable analogue of (SV1) provides open bands around $C^1$-near unit circles about $x$ and $y$, coloured almost everywhere by the same three outsider colours, with $1 < |x-y| < 2$. Then the outsider colours form a weak 3-colouring of a band around a near bicycle. The Main Theorem says this is impossible. So the Main Theorem is exactly the measurable form of (SV3).

The other steps remain. Measurable versions of (SV1) and (SV2) appear to be within reach, at least away from degenerate "pinch" configurations. The real obstruction is the *supply* of trichromatic points in a weak measurable 6-colouring, together with step (SV4). These questions are global: they must use the hypothesis of six colours on the whole plane. This paper says nothing about them. They remain open, and so does the Hadwiger–Nelson problem.

1.4. **Strategy of the proof.** We write $\omega(z)$ for the unit circle about $z$. A *partner* of $y$ is a point at distance exactly 1 from $y$.

*Step counts off $\Sigma_3$.* Let $f$ be a weak 3-colouring of $\Omega$, and let $H_c$ be the essential support of the colour class $A_c$. On $\Omega \setminus \Sigma_3$, where $\Sigma_3 = H_1 \cap H_2 \cap H_3$, the colouring is locally 2-valued. Locally 2-valued maps into $\mathbb{Z}/3$ lift to $\mathbb{Z}$ uniquely up to $3\mathbb{Z}$. This gives a covering space and integer step counts, with no regularity at all (Section 4).

*Index estimate for pair paths.* Ordered pairs of distinct colours form a 6-cycle. A two-colour rigidity lemma (Lemma 2.5) shows that the pair map $(f(x), f(y))$ on unit pairs is locally confined to a contractible arc of this cycle. Hence the classical estimate holds for arbitrary continuous paths of unit pairs $(x(t), y(t))$ off $\Sigma_3$: the step counts of $x(\cdot)$ and $y(\cdot)$ differ by at most one (Proposition 4.3).

*$\Sigma_3$ is small.* The structural core is the following.

- A *shell lemma* (Lemma 5.1) concerns points where $A_a$ meets the support of $A_b$ in positive measure. Such points force thin shells of the third colour along their unit circles, and force the third colour to vanish near their non-degenerate "second partners".
- Consequences: $\Sigma_3$ has empty interior (Theorem 5.3). Next, an open "$F$-region", where exactly two colours are dense, forbids two-colour accumulation continua near its partners (cone lemma, Theorem 6.6). Finally, $\Sigma_3$ contains no non-degenerate continuum in the band (Theorem 6.8).

*Removability.* Unit translation preserves monodromy. Every point of $\Sigma_3$ has a partner ball free of $\Sigma_3$. So the step-count covering has trivial monodromy around $\Sigma_3$, and it extends across the 0-dimensional set $\Sigma_3$ (Section 7).

*Geometry.* The unit-distance correspondence between two near-unit circles is a regular level set in a torus, and it is stable under $C^1$-perturbation (Section 3). For exact circles we describe it completely, including the fold that appears for $|u - v| > \sqrt{3}$, where one partner path overshoots (Proposition 3.1). Only homotopy classes enter the proof.

*Conclusion.* The index estimate along one closed path of unit pairs gives $|s| \leqslant 1$ on a 120° arc. The 60°-chain gives $|s| \geqslant 2$ on the same arc (Section 8).

## 2. Preliminaries

### 2.1. Notation.

- $|E|$ is Lebesgue measure in $\mathbb{R}^2$, and $\mathcal{H}^1$ is length.
- $B(x,r)$ is an open disc, and $\mathbf{e}(\theta) = (\cos\theta, \sin\theta)$.
- $\omega(z) = \{z + \mathbf{e}(\theta)\}$ is the unit circle about $z$, and $N_\delta(X)$ is the open $\delta$-neighbourhood of $X$.
- Colours are $1, 2, 3$. Whenever $a, b, c$ appear together, $\{a, b, c\} = \{1, 2, 3\}$.
- All constants $C, c$ are absolute unless indicated otherwise.

### 2.2. Weak colourings of open sets.

**Definition 2.1.** Let $\Omega \subset \mathbb{R}^2$ be open, and let $\mu$ be the measure on

$$M = \{(x,y) \in \Omega^2 : |x - y| = 1\}$$

obtained by pushing forward $dz\, d\theta$ under $(z,\theta) \mapsto (z, z+\mathbf{e}(\theta))$. A *weak (measurable) $k$-colouring* of $\Omega$ is a Lebesgue measurable $f\colon \Omega \to \{1,\dots,k\}$ such that the set of pairs $(x,y) \in M$ with $f(x) = f(y)$ is $\mu$-null. The colour classes are $A_i = f^{-1}(i)$.

Basic facts about this definition:

- The restriction of a weak measurable colouring of $\mathbb{R}^2$ in the sense of [5, Def. 1.2] to an open set is a weak colouring of that set.
- The restriction of a weak colouring to an open subset is again weak.
- The map $(z,\theta) \mapsto (z+\mathbf{e}(\theta), \theta+\pi)$ preserves $dz\, d\theta$. So $\mu$ is invariant under $(x,y) \mapsto (y,x)$.
- If $E \subset \Omega$ is null, then $\{(x,y) \in M : x \in E \text{ or } y \in E\}$ is $\mu$-null.
- $M$ is a 3-manifold, homeomorphic to an open subset of $\Omega \times S^1$ via either coordinate, and $\mu$ is equivalent to its volume measure.

**Lemma 2.2** (Crofton identity)**.** *For measurable $R, S \subset \mathbb{R}^2$,*

$$\int_R \mathcal{H}^1(\omega(w) \cap S)\, dw = \int_S \mathcal{H}^1(\omega(y) \cap R)\, dy.$$

*Proof.* Both sides equal the $dw\, d\theta$-measure of $\{(w,\theta) : w \in R,\ w + \mathbf{e}(\theta) \in S\}$. The measure-preserving involution above exchanges the roles of $R$ and $S$. □

We repeatedly use the following elementary bound. If $D \subset \mathbb{R}^2$ has diameter $\leqslant 2r$ with $r \leqslant 1/4$, then $\mathcal{H}^1(\omega(y) \cap D) \leqslant 4r$ for every $y$.

**Lemma 2.3** (Basic properties)**.** *Let $f$ be a weak 3-colouring of $\Omega$.*

(W1) *For a.e. $z \in \Omega$, we have $f \neq f(z)$ at $\mathcal{H}^1$-a.e. point of $\omega(z) \cap \Omega$.*
(W2) *Let $T, S \subset \Omega$ be measurable and $c$ a colour. For a.e. $y \in T \cap A_c$, $\mathcal{H}^1(\omega(y) \cap S \cap A_c) = 0$.*

*Proof.* (W1) is Fubini's theorem for $dz\, d\theta$. (W2) is a special case of (W1). □

### 2.3. Lebesgue points and essential supports.

A point $x \in \Omega$ is a *Lebesgue point* of $f$ if

$$|A_c \cap B(x,r)|/|B(x,r)| \to 1 \quad \text{for some (unique) colour } c.$$

By Lebesgue's theorem, a.e. point is a Lebesgue point, and $c = f(x)$ for a.e. $x$. Redefining $f$ on a null set does not affect weakness. Henceforth we assume that $f(x) = c$ at every Lebesgue point $x$.

The *essential support* of $A_c$ is the relatively closed set

$$H_c = \{x \in \Omega : |A_c \cap B(x,r)| > 0 \text{ for all } r > 0\}.$$

We also write

$$K(x) = \{c : x \in H_c\}, \qquad \Sigma_3 = H_1 \cap H_2 \cap H_3.$$

By Lindelöf's theorem, $A_c \setminus H_c$ is null. Hence $H_1 \cup H_2 \cup H_3 = \Omega$: a point outside all $H_c$ would have a neighbourhood in which every colour class is null. So $K(x) \neq \emptyset$ for every $x \in \Omega$. Moreover, every $x$ has a disc $B(x,r)$ on which $f \in K(x)$ almost everywhere.

For a measurable set $E$, $\operatorname{supp} E$ denotes the set of $x$ with $|E \cap B(x,r)| > 0$ for all $r > 0$.

**Lemma 2.4.** *Let $f$ be a weak 3-colouring of $\Omega$.*

(W3) *Lebesgue points $x, y \in \Omega$ with $|x-y| = 1$ have $f(x) \neq f(y)$.*
(W4) *If $K(q) = \{c\}$ and $x \in \Omega$ is a partner of $q \in \Omega$, then $c \notin K(x)$.*

*Proof.* (W3) Suppose $f(x) = f(y) = c$, and let $r \to 0$. Put $D_r = \{(z,\theta) : z \in B(x, r/2),\ z + \mathbf{e}(\theta) \in B(y,r)\}$. Then $|D_r| \geqslant c_0 r^3$, because for each $z$ the admissible $\theta$ form an arc of length $\geqslant r/2$. We bound the parts of $D_r$ where a coordinate misses $A_c$:

- The part with $z \notin A_c$ has measure $\leqslant 2\pi |B(x,r) \setminus A_c| = o(r^2) \cdot r$. This uses that the $\theta$-arc has length $\leqslant Cr$.
- By the involution, the part with $z + \mathbf{e}(\theta) \notin A_c$ has measure $\leqslant Cr\,|B(y,r) \setminus A_c| = o(r^3)$.

So $D_r$ contains monochromatic pairs of positive measure, which contradicts weakness.

(W4) There is $r > 0$ such that $B(q,r) \subset \Omega$, $B(x,r) \subset \Omega$, and $f = c$ a.e. on $B(q,r)$. By (W2), $\mathcal{H}^1(\omega(z) \cap B(x, r/2) \cap A_c) = 0$ for a.e. $z \in B(q,r)$. Lemma 2.2 then gives

$$\int_{A_c \cap B(x,r/2)} \mathcal{H}^1(\omega(y) \cap B(q,r))\, dy = 0.$$

For $y \in B(x, r/2)$ the circle $\omega(y)$ passes within $r/2$ of $q$, so the integrand is positive. Hence $|A_c \cap B(x, r/2)| = 0$. □

2.4. **Two-colour rigidity.**

**Lemma 2.5** (Lemma X)**.** *Let $|p-q| = 1$, let $0 < r < 1/2$, and suppose $B(p,r), B(q,r) \subset \Omega$. Suppose also that $f \in \{i,j\}$ a.e. on both discs. Then $f$ is a.e. constant on $B(q, r/4)$ and on $B(p, r/4)$, with different constants.*

*Proof. Step 1.* By (W1), for a.e. $z \in B(p,r)$, $f$ avoids $f(z)$ a.e. on $\omega(z) \cap B(q,r)$. So $f$ is a.e. constant on that arc.

*Step 2.* Define

$$\Phi(z, \theta_1, \theta_2) = (z + \mathbf{e}(\theta_1),\ z + \mathbf{e}(\theta_2)) \in \mathbb{R}^4.$$

Subtracting the first two rows from the last two in the Jacobian matrix shows $|\det D\Phi| = |\sin(\theta_2 - \theta_1)|$. So $\Phi$ is a local diffeomorphism on

$$D = \{(z, \theta_1, \theta_2) : z \in B(p,r),\ z + \mathbf{e}(\theta_i) \in B(q,r),\ 0 < |\theta_1 - \theta_2| < \pi\}.$$

By Step 1 and Fubini, $\Phi^{-1}(\mathrm{Bad}) \cap D$ is null, where $\mathrm{Bad} = \{(y_1, y_2) : f(y_1) \neq f(y_2)\}$. Local diffeomorphisms map null sets to null sets, so $\mathrm{Bad} \cap \Phi(D)$ is null.

*Step 3.* Let $t$ be a unit vector orthogonal to $p - q$, and let $C$ be the open double cone of directions within angle $r/8$ of $\pm t$. We claim that every pair $(y_1, y_2)$ with $y_i \in B(q, r/4)$ and $y_2 - y_1 \in C$ lies in $\Phi(D)$.

Put $s = |y_2 - y_1| < r/2$. Let $n$ be the unit normal to $y_2 - y_1$ with $n \cdot (p - q) > 0$. Then $|n - (p-q)| \leqslant r/8$. The centre $z = \frac{y_1 + y_2}{2} + \sqrt{1 - s^2/4}\, n$ of a unit circle through $y_1, y_2$ satisfies

$$|z - p| \leqslant r/4 + s^2/4 + r/8 < r.$$

*Step 4.* By Fubini, for a.e. $y_1 \in B(q, r/4)$ we have $f = f(y_1)$ a.e. on $(y_1 + C) \cap B(q, r/4)$.

Fix $x_0 \in B(q, r/4)$ and choose $\lambda > 0$ such that $B(x_0, 4\lambda) \subset B(q, r/4)$. Let $y_1, y_1'$ be two such good points in $B(x_0, \lambda \sin(r/8)/4)$. Then the disc $B(x_0 + 2\lambda t, \lambda \sin(r/8)/4)$ lies in both $(y_1 + C) \cap B(q, r/4)$ and $(y_1' + C) \cap B(q, r/4)$. Hence $f(y_1) = f(y_1')$.

So $f$ is locally a.e. constant on the connected set $B(q, r/4)$, hence a.e. constant there. The same argument with the roles of $p$ and $q$ exchanged works at $p$. Finally, (W3) at Lebesgue points shows that the two constants differ. □

2.5. **A thin-intersection estimate.**

**Lemma 2.6.** *There are absolute constants $c_*, C_* > 0$ with the following property. Let $y, y'$ satisfy $0 < |y - y'| < 2$, let $X$ be a point of $\omega(y) \cap \omega(y')$, and let $\sigma = |\sin \angle(X - y, X - y')| > 0$. Let $0 < s \leqslant c_* \sigma^2$, and let $R$ be the connected component containing $X$ of*

$$\{w : \ ||w - y| - 1| < s/2, \ ||w - y'| - 1| < s/2\}.$$

*Then*

- $s^2/(2\sigma) \leqslant |R| \leqslant 2s^2/\sigma$;
- $R \subset B(X, C_* s/\sigma)$;
- *every $w \in R$ has $\mathcal{H}^1(\omega(w) \cap B(y, s)) \geqslant s$ and $\mathcal{H}^1(\omega(w) \cap B(y', s)) \geqslant s$.*

*Proof.* Consider the map $\Psi(w) = (|w - y|, |w - y'|)$. Its differential at $X$ has rows equal to the unit vectors $X - y$ and $X - y'$, so $|\det D\Psi(X)| = \sigma$. Moreover $|D^2\Psi| \leqslant 4$ on $B(X, 1/2)$.

By the quantitative inverse function theorem, $\Psi$ is a diffeomorphism from a neighbourhood of $X$ onto a square of side $\geqslant c\sigma^2$ around $(1, 1)$. On the preimage of the square $(1 - s/2, 1 + s/2)^2$, which is $R$, $D\Psi$ differs from $D\Psi(X)$ by at most $O(s/\sigma)$, and this is $\leqslant \sigma/10$ when $c_*$ is small. This gives the area bounds and the diameter bound.

For the last item, $\omega(w)$ passes within $s/2$ of $y$, and $\omega(w)$ deviates from its tangent line by $O(s^2)$ inside $B(y, s)$. So the chord has length $\geqslant \sqrt{3}s - O(s^2) \geqslant s$. The same holds at $y'$. □

## 3. The geometry of two unit circles

Paths in a Jordan curve $\Gamma$ that is a radial graph about a centre $o$ are handled through the angular coordinate $\theta = \arg(\cdot - o)$. Such a path is determined up to homotopy rel endpoints by the total change of a continuous lift of $\theta$ along it. We call this total change the *angular displacement*. Throughout,

$$\mathbb{T}^2 = (\mathbb{R}/2\pi\mathbb{Z})^2$$

is the flat torus.

3.1. **Exact circles.** Fix $1 < d < 2$ and put $u = 0$, $v = (d, 0)$, and

$$\theta_0 = \arccos(d/2) \in (0, \tfrac{\pi}{3}), \qquad h = \sin\theta_0, \qquad p = (d/2, h) = \mathbf{e}(\theta_0).$$

So $p$ is the upper point of $T(u) \cap T(v)$, and $p = v + \mathbf{e}(\beta_p)$ with $\beta_p = \pi - \theta_0$. Define

$$t_e = \arccos\frac{d^2 - 3}{2d}, \qquad \lambda = \arccos\frac{d^2 + 3}{4d}.$$

Note that $\frac{d^2-3}{2d} \in (-1, \frac{1}{4})$, so $t_e \in (\arccos\frac{1}{4}, \pi)$. Also $\frac{d^2+3}{4d} \in [\frac{\sqrt{3}}{2}, 1)$ by the AM–GM inequality, so $\lambda \in (0, \frac{\pi}{6}]$.

For $t \in [\theta_0, t_e]$ put $x(t) = \mathbf{e}(t)$ and

$$r(t) = |x(t) - v| = \sqrt{1 + d^2 - 2d\cos t}.$$

Since $\frac{d}{dt}r^2 = 2d\sin t > 0$ on $(0, \pi)$, $r$ increases from $r(\theta_0) = 1$ to $r(t_e) = 2$. Let

$$\beta(t) = \arg(x(t) - v) \in (0, \pi),$$

which is continuous because $\sin t > 0$. Write $e_1 = x(t_e)$ for the point of the upper half of $T(u)$ with $|e_1 - v| = 2$, and $m_1 = \frac{e_1+v}{2}$.

**Proposition 3.1** (Exact geometry)**.** *In the setting above, the following hold.*

(a) (Partners.) *For $t \in [\theta_0, t_e]$, the partners of $x(t)$ on $T(v)$ are exactly $y_\pm(t) = v + \mathbf{e}(\varphi_\pm(t))$, where*

$$\varphi_\pm(t) = \beta(t) \pm \arccos \frac{r(t)}{2}.$$

*They coincide only at $t = t_e$, where $y_\pm = m_1$. At $t = \theta_0$ they equal $R^v_{\pm\pi/3}p = v + \mathbf{e}(\beta_p \pm \frac{\pi}{3})$, where $R^v_\psi$ is rotation about $v$ by $\psi$.*

(b) (Homotopy class.) *For $s \in [-1, 1]$ let $\tau(s) = \theta_0 + (1 - |s|)(t_e - \theta_0)$ and*

$$\Lambda_0(s) = \big(x(\tau(s)),\, y_{\operatorname{sgn} s}(\tau(s))\big),$$

*with either sign at $s = 0$. Then $\Lambda_0$ is a continuous path of unit pairs in $T(u) \times T(v)$ from $(p, R^v_{-\pi/3}p)$ to $(p, R^v_{\pi/3}p)$. Its first projection is a null-homotopic loop at $p$. Its second projection has angular displacement exactly $2\pi/3$, so it is homotopic rel endpoints to the counterclockwise $120^\circ$ arc $\eta_0$ of $T(v)$ centred at $p$.*

(c) (Position of $m_1$.) *The angle at $v$ from $p$ to $m_1$ is $\beta(t_e) - \beta_p = \theta_0 - \lambda$. It satisfies $|\theta_0 - \lambda| < \frac{\pi}{3}$. It is positive for $d < \sqrt{3}$, zero for $d = \sqrt{3}$ (then $m_1 = p$), and negative for $d > \sqrt{3}$. In particular $m_1$ lies in the interior of $\eta_0$.*

(d) (Monotonicity and fold.) *As $x$ moves from $e_1$ to $p$ (that is, as $t$ decreases), $\varphi_+$ is strictly increasing, from $\beta(t_e)$ to $\beta_p + \frac{\pi}{3}$.*

- *If $d \leqslant \sqrt{3}$, $\varphi_-$ is strictly decreasing.*
- *If $d > \sqrt{3}$, $\varphi_-$ has exactly one critical point in $(\theta_0, t_e)$, namely $t = \lambda$, where $x(\lambda) = \frac{u+f_1}{2}$ and $f_1 = v + \mathbf{e}(\pi - t_e)$ is the point of $T(v)$ with $|f_1 - u| = 2$. There $\varphi_-$ attains its minimum $\pi - t_e$. Thus $y_-$ overshoots $R^v_{-\pi/3}p$ by the angle $t_e - \theta_0 - \frac{\pi}{3} > 0$ and then returns.*

(e) (Regularity.) *The set $Z_0 = \{(t, \varphi) \in \mathbb{T}^2 : |\mathbf{e}(t) - v - \mathbf{e}(\varphi)| = 1\}$ is the zero set of $G_0(t, \varphi) = |\mathbf{e}(t) - v - \mathbf{e}(\varphi)|^2 - 1$. The value $0$ is a regular value of $G_0$, so $Z_0$ is a compact smooth 1-submanifold of $\mathbb{T}^2$ containing the path $\Lambda_0$.*

*Proof.* (a) Write $x - v = r\mathbf{e}(\beta)$. Then

$$|v + \mathbf{e}(\varphi) - x|^2 = 1 + r^2 - 2r\cos(\varphi - \beta).$$

This equals 1 iff $\cos(\varphi - \beta) = r/2$, which gives the formula. The two solutions coincide iff $r = 2$. At $t = \theta_0$ we have $r = 1$, $\arccos \frac{1}{2} = \frac{\pi}{3}$ and $\beta = \beta_p$.

(b) Continuity at $s = 0$ holds because $\varphi_+(t_e) = \varphi_-(t_e)$. The first projection runs along the arc $x([\theta_0, t_e])$ and back, so it is null-homotopic. The second projection has the continuous lift

$$s \mapsto \varphi_{\operatorname{sgn} s}(\tau(s)),$$

whose total change is $\varphi_+(\theta_0) - \varphi_-(\theta_0) = 2\arccos \frac{1}{2} = \frac{2\pi}{3}$. The arc $\eta_0$ has the lift $\beta_p + \psi$, $\psi \in [-\frac{\pi}{3}, \frac{\pi}{3}]$, with the same endpoints.

(c) We have $e_1 - v = 2\mathbf{e}(\beta(t_e))$. Its first coordinate is

$$\cos t_e - d = \frac{d^2 - 3}{2d} - d = -\frac{d^2 + 3}{2d}.$$

Hence $\cos \beta(t_e) = -\frac{d^2+3}{4d} = \cos(\pi - \lambda)$, and $\beta(t_e) = \pi - \lambda$ because $\beta \in (0, \pi)$. So $\beta(t_e) - \beta_p = \theta_0 - \lambda$.

Since $\theta_0 \in (0, \frac{\pi}{3})$ and $\lambda \in (0, \frac{\pi}{6}]$, we get $|\theta_0 - \lambda| < \frac{\pi}{3}$. Finally,

$$\theta_0 > \lambda \iff \cos\theta_0 < \cos\lambda \iff \frac{d}{2} < \frac{d^2+3}{4d} \iff d^2 < 3.$$

(d) Write $G_0(t,\varphi) = |x(t) - y(\varphi)|^2 - 1$ with $y(\varphi) = v + \mathbf{e}(\varphi)$. Then:

- $\partial_\varphi G_0 = -2(x-y)\cdot y'(\varphi)$ vanishes iff $x$ lies on the line through $v$ and $y$. Since $|x-y| = 1 = |y-v|$, this means $|x-v| \in \{0,2\}$.
- Likewise $\partial_t G_0 = 0$ iff $|y-u| \in \{0,2\}$.

On $(\theta_0, t_e)$ we have $r \in (1,2)$, so $\partial_\varphi G_0 \neq 0$ along both branches. Implicit differentiation of $G_0(t, \varphi_\pm(t)) = 0$ shows that $\varphi'_\pm(t) = 0$ iff the partner $y_\pm(t)$ satisfies $|y_\pm(t) - u| = 2$ (the value 0 is impossible since $|u-v| > 1$).

The set $J = \{y \in T(v) : |y-u| \leqslant 2\}$ is the arc $\{v + \mathbf{e}(\varphi) : \varphi \in [\pi - t_e, \pi + t_e]\}$. Indeed,

$$|v + \mathbf{e}(\varphi)|^2 = 1 + d^2 + 2d\cos\varphi \leqslant 4 \iff \cos\varphi \leqslant \frac{3-d^2}{2d} = \cos(\pi - t_e).$$

Its endpoints $f_1 = v + \mathbf{e}(\pi - t_e)$ and $\bar{f}_1$ are the only points of $T(v)$ at distance 2 from $u$. The unique partner of $f_1$ on $T(u)$ is $\frac{f_1}{2}$. A direct computation gives

$$\frac{f_1}{2} = \Big(\frac{d - \cos t_e}{2}, \frac{\sin t_e}{2}\Big) = \mathbf{e}(\lambda),$$

because $\frac{d-\cos t_e}{2} = \frac{d^2+3}{4d}$. Since $\bar{f}_1/2$ lies in the lower half-plane, an interior critical point of $\varphi_\pm$ on $(\theta_0, t_e)$ occurs iff $\lambda \in (\theta_0, t_e)$. As $\lambda \leqslant \frac{\pi}{6} < t_e$, this happens iff $\lambda > \theta_0$, that is, iff $d > \sqrt{3}$ by (c). At most one branch is critical there, and only at $t = \lambda$.

Suppose $d \leqslant \sqrt{3}$. Then both $\varphi_\pm$ are strictly monotone on $[\theta_0, t_e]$. Their directions follow from the endpoint values: $\varphi_\pm(t_e) = \beta(t_e) = \beta_p + (\theta_0 - \lambda)$, $\varphi_\pm(\theta_0) = \beta_p \pm \frac{\pi}{3}$, and $|\theta_0 - \lambda| < \frac{\pi}{3}$.

Suppose $d > \sqrt{3}$. Then $t_e < \frac{2\pi}{3}$, since $\frac{d^2-3}{2d} > -\frac{1}{2}$. We first show that $f_1$ is the partner $y_-(\lambda)$, not $y_+(\lambda)$. If $f_1 = y_+(\lambda)$, then $y_-(\lambda)$ would lie clockwise of $f_1$ at an angle in $(0, \frac{2\pi}{3}]$. It would then lie outside $J$, because the complementary arc of $J$ is the open arc of angular length $2\pi - 2t_e > \frac{2\pi}{3}$ adjacent to $f_1$ on its clockwise side. But every partner of a point of $T(u)$ lies in $J$, which is a contradiction.

So $\varphi_+$ has no critical point and is strictly increasing as $t$ decreases, by the endpoint values. For $\varphi_-$, note that $\varphi_- \in (-\frac{\pi}{3}, \pi)$ and $y_- \in J$. Since $t_e - \pi < -\frac{\pi}{3}$, this forces $\varphi_-(t) \in [\pi - t_e, \pi)$. The value $\pi - t_e$ is attained exactly at the unique critical point $t = \lambda$, so it is the strict minimum. The overshoot is

$$\varphi_-(\theta_0) - \min\varphi_- = \Big(\beta_p - \frac{\pi}{3}\Big) - (\pi - t_e) = t_e - \theta_0 - \frac{\pi}{3},$$

which is positive because the minimum is attained at an interior point and $\varphi_-$ is strictly monotone on both sides of it.

(e) A critical point of $G_0$ on $Z_0$ needs $|x-v| \in \{0,2\}$ and $|y-u| \in \{0,2\}$ simultaneously. The value 0 is excluded for both, since $|u-v| > 1$. So $|x-v| = 2$ and $|y-u| = 2$, with $|x-u| = |y-v| = |x-y| = 1$. Then $y = 2x - u$, so $x$ is the midpoint of $u$ and $y$. Likewise $x = 2y - v$, so $y$ is the midpoint of $x$ and $v$. Hence $u, x, y, v$ are collinear with consecutive distances 1, which forces $d = 3$. This contradicts $d < 2$. □

**Remark 3.2.** For $d = 1.9$ and $d = 1.99$, the overshoot in (d) is $2.57°$ and $10.31°$; it tends to $\arccos\frac{1}{4} - 60° \approx 15.5°$ as $d \to 2$. At $d = 1.99$ the angle $\theta_0 - \lambda$ is $-23.3°$. We checked the formulas of (a), (c) and (d) numerically on fine grids for many values of $d$. These checks were not used in the proof. Only (a), (b) and (e) are used below. Parts (c) and (d) explain why monotonicity

of the partner paths, which earlier formulations of the index argument implicitly relied on, fails for $d > \sqrt{3}$. They also explain why it is irrelevant: only homotopy classes matter.

### 3.2. **Stability of regular level sets.**

**Lemma 3.3.** *Let $G_0 \in C^1(\mathbb{T}^2)$ have $0$ as a regular value, with $Z_0 = G_0^{-1}(0)$ a compact $C^1$ submanifold. Then there are $\kappa_1, C_1 > 0$ with the following property. Let $G \in C^1(\mathbb{T}^2)$ satisfy $\|G - G_0\|_{C^1} < \kappa_1$, where $\|g\|_{C^1} = \sup|g| + \sup|\nabla g|$. Then there is a continuous function $\varsigma\colon Z_0 \to \mathbb{R}$ with $|\varsigma| \leqslant C_1 \sup|G - G_0|$ such that*

$$h\colon Z_0 \to G^{-1}(0), \qquad h(z) = z + \varsigma(z)n(z), \qquad n = \frac{\nabla G_0}{|\nabla G_0|},$$

*is a homeomorphism.*

*Proof.* Let $m = \min_{Z_0} |\nabla G_0| > 0$. By compactness there is $\varepsilon_0 > 0$ with two properties:

- $\Psi(z, s) = z + sn(z)$ is a homeomorphism of $Z_0 \times [-\varepsilon_0, \varepsilon_0]$ onto a closed neighbourhood $U$ of $Z_0$;
- $\nabla G_0(\Psi(z, s)) \cdot n(z) \geqslant m/2$ on $Z_0 \times [-\varepsilon_0, \varepsilon_0]$.

Then $\pm G_0(\Psi(z, \pm\varepsilon_0)) \geqslant m\varepsilon_0/2$. Let $m' = \min\{|G_0(w)| : w \in \overline{\mathbb{T}^2 \setminus U}\} > 0$, and put

$$\kappa_1 = \min\Big(\frac{m}{4},\ m',\ \frac{m\varepsilon_0}{4}\Big).$$

Suppose $\|G - G_0\|_{C^1} < \kappa_1$. Then $G$ has no zero outside $U$. For each $z \in Z_0$, the function $s \mapsto G(\Psi(z, s))$ has derivative $\geqslant m/2 - \kappa_1 \geqslant m/4$, and it has opposite signs at $s = \pm\varepsilon_0$. So it has a unique zero $\varsigma(z)$, with

$$|\varsigma(z)| \leqslant \frac{4|G(z)|}{m} = \frac{4|G(z) - G_0(z)|}{m}.$$

A unique zero of a continuous family of strictly increasing functions depends continuously on the parameter. Thus $h$ is a continuous bijection of the compact space $Z_0$ onto $G^{-1}(0)$, hence a homeomorphism. □

### 3.3. **Near bicycles.**

**Proposition 3.4** (Geometry of near bicycles)**.** *For every $d \in (1, 2)$ there is $\kappa_0(d) \in (0, \frac{1}{8}]$ with the following property. Let $(\Gamma_u, \Gamma_v)$ be a $\kappa$-near bicycle with $\kappa \leqslant \kappa_0$, and write $Y(\varphi) = v + \rho_v(\varphi)\mathbf{e}(\varphi)$. Then there are points $p \in \Gamma_u \cap \Gamma_v$ and $q_\pm \in \Gamma_v$ with $|p - q_\pm| = 1$, and lifts $\varphi_- < \varphi_p < \varphi_+$ of their angles about $v$ with $\varphi_+ - \varphi_- < \pi$, with the following properties.*

(i) *Let $\eta_1$ and $\eta_2$ be the arcs $Y([\varphi_-, \varphi_p])$ and $Y([\varphi_p, \varphi_+])$, oriented by increasing $\varphi$, and let $\eta = \eta_1 \cdot \eta_2$. There is a continuous path $\Lambda$ in*

$$Z_\Gamma = \{(x, y) \in \Gamma_u \times \Gamma_v : |x - y| = 1\}$$

*from $(p, q_-)$ to $(p, q_+)$ such that $\pi_1\Lambda$ is a loop at $p$ that is null-homotopic in $\Gamma_u$, and $\pi_2\Lambda$ is homotopic to $\eta$ rel endpoints in $\Gamma_v$.*

(ii) *There is a continuous path $(z(s), z'(s))_{s\in[0,1]}$ in $\Gamma_v \times \Gamma_v$ with $|z(s) - z'(s)| = 1$, from $(q_-, p)$ to $(p, q_+)$, such that $z = \eta_1$ (after reparametrisation) and $z'$ is homotopic to $\eta_2$ rel endpoints in $\Gamma_v$.*

*Proof.* Put $X(t) = u + \rho_u(t)\mathbf{e}(t)$ and $X_0(t) = u + \mathbf{e}(t)$, and similarly $Y_0(\varphi) = v + \mathbf{e}(\varphi)$. We have

$$|X - X_0| = |\rho_u - 1| \leqslant \kappa, \qquad |X' - X_0'| = |\rho_u'\mathbf{e} + (\rho_u - 1)\mathbf{e}^\perp| \leqslant 2\kappa,$$

and the same bounds hold for $Y$. All points involved are within distance 5 of each other. Hence

$$G(t, \varphi) = |X(t) - Y(\varphi)|^2 - 1 \quad \text{satisfies} \quad \|G - G_0\|_{C^1} \leqslant 40\kappa,$$

where $G_0$ is the function of Proposition 3.1(e) (translated so that $u = 0$).

*The point $p$.* Consider $k(t) = |X(t) - v| - \rho_v(\arg(X(t) - v))$ on a fixed neighbourhood $I$ of $\theta_0$; there $X(t) - v$ stays away from 0. A point $X(t)$ lies on $\Gamma_v$ iff $k(t) = 0$. For the exact circles,

$$k_0(t) = |\mathbf{e}(t) - v| - 1, \qquad k_0(\theta_0) = 0, \qquad k_0'(\theta_0) = d \sin\theta_0 > 0.$$

Since $\|k - k_0\|_{C^1(I)} \leqslant C(d)\kappa$, for small $\kappa$ there is a unique zero $t_p \in I$ with $|t_p - \theta_0| \leqslant C(d)\kappa$. Put $p = X(t_p)$ and $\varphi_p = \arg(p - v)$, lifted near $\beta_p$.

*The points $q_\pm$.* We have $\partial_\varphi G_0(\theta_0, \beta_p \pm \frac{\pi}{3}) \neq 0$, because $|p_0 - v| = 1 \notin \{0, 2\}$, where $p_0 = \mathbf{e}(\theta_0)$ is the exact point. So, for small $\kappa$, $\varphi \mapsto G(t_p, \varphi)$ has a unique zero $\varphi_\pm$ within a fixed distance of $\beta_p \pm \frac{\pi}{3}$, and $|\varphi_\pm - (\beta_p \pm \frac{\pi}{3})| \leqslant C(d)\kappa$. Put $q_\pm = Y(\varphi_\pm)$. Then $\varphi_- < \varphi_p < \varphi_+$ and $\varphi_+ - \varphi_- < \pi$.

*Proof of (i).* Apply Lemma 3.3 to $G_0$, which is legitimate by Proposition 3.1(e). For $40\kappa < \kappa_1$ we get a homeomorphism $h\colon Z_0 \to Z_\Gamma$ (in coordinates $(t, \varphi)$) that moves points by at most $C(d)\kappa$. The points $z_\pm = h^{-1}(t_p, \varphi_\pm)$ of $Z_0$ lie within $C(d)\kappa$ of the endpoints $(\theta_0, \beta_p \pm \frac{\pi}{3})$ of $\Lambda_0$.

Since $Z_0$ is a compact embedded curve, small balls around these endpoints meet $Z_0$ in single arcs. So for small $\kappa$ we can extend $\Lambda_0$ at both ends by arcs $a_\pm$ of $Z_0$ of diameter $\leqslant C(d)\kappa$, obtaining a path $\Lambda_0' = a_- \cdot \Lambda_0 \cdot a_+$ in $Z_0$ from $z_-$ to $z_+$. Set $\Lambda = h \circ \Lambda_0'$. This is a path in $Z_\Gamma$ from $(t_p, \varphi_-)$ to $(t_p, \varphi_+)$, that is, from $(p, q_-)$ to $(p, q_+)$.

Next we compare angular displacements. Two paths in $\mathbb{T}^2$ that stay pointwise within distance $\epsilon < \pi$ of each other have lifts that stay within $\epsilon$ of each other. So the displacement of each coordinate along $\Lambda$ differs by at most $C(d)\kappa$ from the displacement along $\Lambda_0$:

- For the $t$-coordinate, the displacement along $\Lambda_0$ is 0. Along $\Lambda$ it lies in $2\pi\mathbb{Z}$, because $\pi_1\Lambda$ is a loop, and it has absolute value $\leqslant C(d)\kappa < 2\pi$. So it is 0, and $\pi_1\Lambda$ is null-homotopic.
- For the $\varphi$-coordinate, the displacement along $\Lambda_0$ is $\frac{2\pi}{3}$ by Proposition 3.1(b). Along $\Lambda$ it equals $\varphi_+ - \varphi_- + 2\pi k$ for some $k \in \mathbb{Z}$, and it lies within $C(d)\kappa$ of $\frac{2\pi}{3}$. Since $\varphi_+ - \varphi_-$ is also within $C(d)\kappa$ of $\frac{2\pi}{3}$, we get $k = 0$. This is exactly the displacement of $\eta$.

*Proof of (ii).* Let $H(\varphi, \varphi') = |Y(\varphi) - Y(\varphi')|^2 - 1$ and $H_0 = 1 - 2\cos(\varphi' - \varphi)$. On the window $W_\varphi = [\varphi + \frac{\pi}{6}, \varphi + \frac{\pi}{2}]$ we have

$$\partial_{\varphi'} H_0 = 2\sin(\varphi' - \varphi) \geqslant 1, \qquad H_0(\varphi, \varphi + \tfrac{\pi}{6}) = 1 - \sqrt{3} < 0, \qquad H_0(\varphi, \varphi + \tfrac{\pi}{2}) = 1.$$

Since $\|H - H_0\|_{C^1} \leqslant C\kappa$, for small $\kappa$ each $\varphi$ has a unique $\psi(\varphi) \in W_\varphi$ with $H(\varphi, \psi(\varphi)) = 0$, and $\psi$ is continuous.

The point $p = Y(\varphi_p)$ is a partner of $q_- = Y(\varphi_-)$, and $\varphi_p - \varphi_-$ is within $C\kappa$ of $\frac{\pi}{3}$, so $\varphi_p \in W_{\varphi_-}$ and $\psi(\varphi_-) = \varphi_p$. Likewise $\psi(\varphi_p) = \varphi_+$. The path $\varphi \mapsto (Y(\varphi), Y(\psi(\varphi)))$, $\varphi \in [\varphi_-, \varphi_p]$, has first projection $\eta_1$. Its second projection has the continuous lift $\psi$ from $\varphi_p$ to $\varphi_+$, so it is homotopic rel endpoints to $\eta_2$. □

**Remark 3.5** (On $C^2$ versus $C^1$, and folds)**.** An earlier formulation of the stability claim asserted that the cross-partner maps are local diffeomorphisms and that the fold at $(e_1, m_1)$ is nondegenerate. The first assertion is false for $d > \sqrt{3}$ by Proposition 3.1(d). Neither assertion is needed.

The proof above uses only that $Z_0$ is a regular level set (Proposition 3.1(e)) and the elementary Lemma 3.3. Both are insensitive to folds of the coordinate projections and require only $C^1$-closeness. In particular the Main Theorem holds for $C^2$-near bicycles, which were the curves considered originally.

3.4. **Room in the band.** Let $(\Gamma_u, \Gamma_v)$ be a $\kappa$-near bicycle with $\kappa \leqslant \kappa_0 \leqslant \frac{1}{8}$, and let $0 < \delta \leqslant \frac{1}{8}$. Put

$$\Omega_0 = N_{\delta/2}(\Gamma_u \cup \Gamma_v), \qquad \Omega_1 = N_\delta(\Gamma_u \cup \Gamma_v).$$

Shrinking $\Omega$ does not affect the conclusion of the Main Theorem, since weak colourings restrict to weak colourings. So we may and do assume $\delta \leqslant \frac{1}{8}$.

**Lemma 3.6** (Room)**.** *Every $y \in \Omega_0$ has a partner in $\Gamma_u \cup \Gamma_v \subset \Omega_0$.*

*Proof.* Say $y \in N_{\delta/2}(\Gamma_u)$. Then $\big||y-u|-1\big| < \kappa + \delta/2 < \frac{1}{4}$. Put $\nu = (y-u)/|y-u|$. The point $y+\nu \in \omega(y)$ is at distance $> \frac{7}{4} > 1+\kappa$ from $u$, so it lies outside $\Gamma_u$. The point $y-\nu$ is at distance $< \frac{1}{4} < 1-\kappa$ from $u$, so it lies inside $\Gamma_u$. Since $\omega(y)$ is connected, it meets $\Gamma_u$. □

We say that $y \in \Omega_1$ has property (BH) if there are $z_1, y_2 \in \Omega_1$ with $|z_1 - y| = |y_2 - z_1| = 1$ and $y_2 \notin \{y, 2z_1 - y\}$.

**Corollary 3.7.** *Every point of $\Omega_0$ has property* (BH)*.*

*Proof.* Take a partner $z_1 \in \Omega_0$ of $y$, which exists by Lemma 3.6. Points of $\omega(z_1)$ close to $y$ but different from $y$ lie in $\Omega_1$. They are not $2z_1 - y$, which is at distance 2 from $y$. □

## 4. Step counts off $\Sigma_3$

Throughout this section $f$ is a weak 3-colouring of an open set $\Omega$, and $\Omega' = \Omega \setminus \Sigma_3$. A *good point* is a Lebesgue point of $f$ that lies in $\Omega'$.

4.1. **The colour covering.** Let $x \in \Omega'$. Then $|K(x)| \leqslant 2$, so there is a disc $B_x \subset \Omega'$ on which $f$ takes at most two values a.e., both in $K(x)$. A *local lift* on $B_x$ is a map $g\colon \{1,2,3\} \supseteq K(x) \to \mathbb{Z}$ with $g(c) \equiv c \pmod 3$ whose values are equal or adjacent integers. It exists, and it is unique up to adding an element of $3\mathbb{Z}$.

If $B_x \cap B_y \neq \emptyset$, then the overlap has positive measure. The colours occurring a.e. on the overlap form a non-empty subset of $K(x) \cap K(y)$. If this subset is a single colour or the full common 2-set, two local lifts that agree on one occurring colour agree on all occurring colours. Hence local lifts glue along overlaps up to constants in $3\mathbb{Z}$. This defines a covering space $\widehat{\Omega}' \to \Omega'$ with fibre $3\mathbb{Z}$ (a regular covering with deck group $3\mathbb{Z}$).

**Definition 4.1.** Let $\alpha\colon [0,1] \to \Omega'$ be a path between good points. Choose a partition $0 = t_0 < \cdots < t_N = 1$ and discs $B_k = B_{x_k}$ with $\alpha([t_k, t_{k+1}]) \subset B_k$, and good points $y_k \in B_{k-1} \cap B_k$ for $0 < k < N$; put $y_0 = \alpha(0)$, $y_N = \alpha(1)$. The *step count* is

$$s(\alpha) = \sum_{k=0}^{N-1} \big(g_k(f(y_{k+1})) - g_k(f(y_k))\big),$$

where $g_k$ is a local lift on $B_k$.

Good points are dense, so such chains exist. The following properties are the standard facts about path lifting in the covering $\widehat{\Omega}'$; we omit the routine verification.

- $s(\alpha)$ does not depend on the choices, and equals the change of the lifted colour along the lift of $\alpha$.
- $s$ is additive under concatenation and invariant under homotopy rel endpoints in $\Omega'$.
- $s(\alpha) \equiv f(\alpha(1)) - f(\alpha(0)) \pmod 3$.
- For a loop $\lambda$ in $\Omega'$ (with any base point) the monodromy of the covering along $\lambda$ is a well defined element $s(\lambda) \in 3\mathbb{Z}$. It depends only on the free homotopy class of $\lambda$ in $\Omega'$, and it agrees with the step count when the base point is good.

No regularity of the colour classes is used. A path may cross infinitely many colour changes (shells, fat interfaces); only the endpoint colours and the topology of $\Omega'$ enter.

4.2. **The pair map.** Let $\mathcal{C}_6$ be the graph whose vertices are the ordered pairs $(i,j)$ of distinct colours, with $(i,j) \sim (i',j')$ when exactly one coordinate agrees. It is a 6-cycle:

$$(1,2) - (1,3) - (2,3) - (2,1) - (3,1) - (3,2) - (1,2).$$

We label its vertices by $\mathbb{Z}/6$ in this order, starting with $v_0 = (1,2)$. Put

$$G_1(n) = 1 + \lfloor n/2 \rfloor, \qquad G_2(n) = 2 + \lceil n/2 \rceil \qquad (n \in \mathbb{Z}).$$

Then $v_{n \bmod 6} = (G_1(n) \bmod 3,\ G_2(n) \bmod 3)$, both $G_1$ and $G_2$ are non-decreasing with increments in $\{0,1\}$, and

$$G_1(n) - G_2(n) = -1 - (n \bmod 2) \in \{-1,-2\}, \qquad G_i(n+6k) = G_i(n) + 3k. \tag{1}$$

For $(x,y) \in M$ put $F(x,y) = (f(x), f(y))$. By weakness, $F \in \mathcal{C}_6$ for $\mu$-a.e. pair. Let $M' = M \cap (\Omega' \times \Omega')$.

**Corollary 4.2** (of Lemma 2.5)**.** *Every $(p,q) \in M'$ has a neighbourhood $U = M \cap (B(p,r) \times B(q,r))$ and a sub-path $\Pi \subset \mathcal{C}_6$ with at most three vertices such that $F \in \Pi$ $\mu$-a.e. on $U$.*

*Proof.* Choose $r$ so small that $f \in K(p)$ a.e. on $B(p,r)$ and $f \in K(q)$ a.e. on $B(q,r)$, and so that Lemma 2.5 applies. Then $F \in (K(p) \times K(q)) \cap \mathcal{C}_6$ a.e. on $U$.

- If $K(p) = K(q) = \{i,j\}$, Lemma 2.5 shows that $F$ is a.e. constant on the smaller neighbourhood with radius $r/4$.
- If $K(p) = \{i,j\}$ and $K(q) = \{i,k\}$, the admissible pairs are $(i,k),(j,k),(j,i)$, a path in $\mathcal{C}_6$.
- If one of $K(p), K(q)$ is a singleton, the admissible pairs form at most an edge. □

**Proposition 4.3** (Index estimate)**.** *Let $\gamma = (\gamma_1, \gamma_2)\colon [0,1] \to M'$ be a continuous path of unit pairs whose endpoints are pairs of good points. Then*

$$|s(\gamma_1) - s(\gamma_2)| \leqslant 1.$$

*If $\gamma$ is a loop based at a pair of good points, then $s(\gamma_1) = s(\gamma_2)$.*

Injectivity of $\gamma_1$ or $\gamma_2$ is not needed.

*Proof.* Cover $\gamma([0,1])$ by neighbourhoods $U_0, \dots, U_{N-1}$ as in Corollary 4.2, with paths $\Pi_k$, such that $\gamma([t_k, t_{k+1}]) \subset U_k$. The intersection of two sub-paths of $\mathcal{C}_6$ with at most three vertices is connected (possibly empty). Pairs of good points have full $\mu$-measure, so we may choose pairs $P_k = (y_k, y_k') \in U_{k-1} \cap U_k$ of good points with $F(P_k)$ in the essential range of $F$ on $U_{k-1} \cap U_k$. Put $P_0 = \gamma(0)$ and $P_N = \gamma(1)$.

Lift $\Pi_0$ to an interval of $\mathbb{Z}$ (via $n \mapsto v_{n \bmod 6}$). Inductively, lift $\Pi_k$ so that the lifts of $\Pi_{k-1}$ and $\Pi_k$ agree on the connected set $\Pi_{k-1} \cap \Pi_k$, which contains $F(P_k)$. This gives integers $L(P_k)$, $0 \leqslant k \leqslant N$, with $v_{L(P_k) \bmod 6} = F(P_k)$.

On $U_k$, the lifted vertices take at most three consecutive values $n$. By the monotonicity of $G_1$, the values $G_1(n)$ are at most two adjacent integers, congruent mod 3 to the first colour. So $c \mapsto G_1$ is a local lift on $B(p_k, r_k)$ in the sense above, and similarly $G_2$ on $B(q_k, r_k)$. The chains $(y_k)$ and $(y_k')$ follow $\gamma_1$ and $\gamma_2$, so

$$s(\gamma_1) = G_1(L(P_N)) - G_1(L(P_0)), \qquad s(\gamma_2) = G_2(L(P_N)) - G_2(L(P_0)).$$

By (1), $s(\gamma_1) - s(\gamma_2) \in \{-1, 0, 1\}$.

If $\gamma$ is a loop, then $F(P_N) = F(P_0)$, so $L(P_N) - L(P_0) = 6k$, and (1) gives $s(\gamma_1) = 3k = s(\gamma_2)$. □

## 5. The shell lemma and the interior of $\Sigma_3$

5.1. **Scale matching.** The following consequence of Egorov's theorem replaces the regularity used in the classical arguments. Let $E \subset A_a$ have positive measure and $\delta > 0$. Then there are $s^* > 0$ and $E' \subset E$ with $|E'| > 0$ such that

$$|A_a \cap B(x,\varrho)| \geqslant (1-\delta)|B(x,\varrho)| \qquad \text{for all } x \in E',\ \varrho \leqslant s^*.$$

Since $x \mapsto |A_a \cap B(x,\varrho)|$ is continuous, the same holds for all $x \in \overline{E'}$.

Every argument below compares a set of density $\geqslant 1-\delta$ with a family of unit circles *at the same scale*, and concludes only that some density is $\leqslant C\delta$. Openness enters only through the supports $H_c$.

5.2. **The shell lemma.** We say that (Fat)$(a,b)$ holds at $y^* \in \Omega$ if $y^* \in \operatorname{supp}(A_a \cap H_b)$. This holds at every interior point of $H_a \cap H_b$, and at a.e. point of $A_a \cap H_b$. It never holds when the topological boundaries of the colour classes are null.

**Lemma 5.1** (Shell lemma)**.** *Assume* (Fat)$(a,b)$ *at* $y^* \in \Omega$*, and let* $c$ *be the third colour.*

(i) *Let* $z_1$ *be a partner of* $y^*$ *such that the arc of* $\omega(y^*)$ *of angular half-width* $\gamma_0 > 0$ *about* $z_1$ *lies in* $\Omega$*. Then* $z_1 \in H_c$*.*

(ii) *Let moreover* $y_2$ *be a partner of* $z_1$ *with* $y_2 \notin \{y^*, 2z_1 - y^*\}$ *and* $B(y_2, L_0) \subset \Omega$ *for some* $L_0 > 0$*. Then* $c$ *is null on a neighbourhood of* $y_2$*.*

*Proof.* Fix $\gamma \in (0, \gamma_0/4)$, $\rho > 0$ and $\delta \in (0, \frac{1}{8})$. Let $\psi_0$ be the direction of $z_1 - y^*$, and $\varphi_0 = \psi_0 - \frac{\pi}{2}$. All constants $C$ are absolute.

*Step 1: a small b-island in an a-sea.* Apply scale matching to $E = A_a \cap H_b \cap B(y^*, \rho)$, which has positive measure. We get $E'$ and $s^* < \rho$. Let $e_0$ be a density point of $E'$. Then $e_0 \in \overline{E'} \subset H_b$, so there is a Lebesgue point $y_1$ of $A_b$ with $|y_1 - e_0| < s^*/10$.

Put $t = \operatorname{dist}(y_1, \overline{E'}) < s^*/10$. Then $t > 0$. Otherwise there are $e \in \overline{E'}$ with $\tau = |e - y_1|$ arbitrarily small; since $B(y_1,\tau) \subset B(e, 2\tau)$ and $A_a$ has density $\geqslant 1-\delta$ in $B(e,2\tau)$, the density of $A_b$ in $B(y_1,\tau)$ would be $\leqslant 4\delta < \frac{1}{2}$. Choose $e \in \overline{E'}$ with $|e - y_1| = t$. Then $A_a$ has density $\geqslant 1-\delta$ in $B(e,4t)$. Finally choose $s_1 \leqslant c_* t^2/4$ (with $c_*$ from Lemma 2.6) such that $A_b$ has density $\geqslant 1-\delta$ in $B(y_1, s_1)$.

*Step 2: a-dense discs at scale $s_1$.* Let

$$Q = \{y_1 + r\mathbf{e}(\varphi) : t \leqslant r \leqslant 2t,\ |\varphi - \varphi_0| < \gamma\} \subset B(e, 3t), \qquad |Q| = 3\gamma t^2.$$

Put $g(x) = |B(x,s_1) \setminus A_a|/(\pi s_1^2)$. Since $B(x,s_1) \subset B(e,4t)$ for $x \in Q$, Fubini gives $\int_Q g \leqslant |B(e,4t) \setminus A_a| \leqslant 16\pi\delta t^2$. So $G = \{x \in Q : g(x) \leqslant \sqrt{\delta}\}$ satisfies $|Q \setminus G| \leqslant 16\pi\sqrt{\delta}\, t^2$.

*Step 3: parallelograms in which c is dense.* For $e'' \in Q$ with $r = |e'' - y_1|$, let $R(e'')$ be the component near $\omega(y_1) \cap \omega(e'')$ on the side of $\psi_0$ of

$$\{w :\ ||w - y_1| - 1| < s_1/2,\ ||w - e''| - 1| < s_1/2\}.$$

The two circles cross there with $\sigma \in [r/2, r]$, so Lemma 2.6 applies with $s = s_1$. Thus $|R(e'')| \geqslant s_1^2/(2r)$, $R(e'')$ lies in a disc of radius $D = C_* s_1/\sigma$, and each $w \in R(e'')$ has chords of length $\geqslant s_1$ in $B(y_1,s_1)$ and in $B(e'',s_1)$.

Call $w \in R(e'')$ *b-bad* if $\mathcal{H}^1(\omega(w) \cap B(y_1,s_1) \cap A_b) = 0$, and *a-bad* if $\mathcal{H}^1(\omega(w) \cap B(e'',s_1) \cap A_a) = 0$. By Lemma 2.2 and the chord bound $\mathcal{H}^1(\omega(y) \cap R(e'')) \leqslant 4D$,

$$|b\text{-bad}| \cdot s_1 \leqslant |B(y_1,s_1) \setminus A_b| \cdot 4D, \qquad |a\text{-bad}| \cdot s_1 \leqslant |B(e'',s_1) \setminus A_a| \cdot 4D.$$

For $e'' \in G$ both are $\leqslant C\sqrt{\delta}\, s_1^2/r \leqslant C\sqrt{\delta}|R(e'')|$. For a point $w$ that is neither $a$-bad nor $b$-bad, $\omega(w)$ meets $A_a$ and $A_b$ in positive length; by (W1), $f(w) = c$ for a.e. such $w$. Hence

$$|R(e'') \setminus A_c| \leqslant C\sqrt{\delta}\,|R(e'')| \qquad (e'' \in G). \tag{2}$$

*Step 4: the shell.* Let

$$N = \{w : \ ||w - y_1| - 1| < s_1/4, \ |\arg(w - y_1) - \psi_0| < \gamma/2\}, \qquad |N| \asymp \gamma s_1.$$

For $w \in N$ put $J(w) = \{e'' \in Q : w \in R(e'')\} = Q \cap \{e'' : ||e'' - w| - 1| < s_1/2\}$. Near $y_1$ the circle $\omega(w)$ is a nearly straight arc through $B(y_1, s_1/4)$ with direction within $\gamma/2$ of $\varphi_0$. It therefore crosses $Q$ radially, and $\frac{1}{2}s_1 t \leqslant |J(w)| \leqslant 2s_1 t$.

Using $\int_N |J(w) \cap S| \, dw = \int_S |R(e'') \cap N| \, de''$ for $S \subset Q$, and $|R(e'')| \leqslant C s_1^2/t$:

- with $S = Q \setminus G$: the set $N_{\mathrm{bad}} = \{w \in N : |J(w) \cap G| < \frac{1}{4}s_1 t\}$ satisfies $|N_{\mathrm{bad}}| \cdot \frac{1}{4}s_1 t \leqslant C\sqrt{\delta}\, t^2 \cdot s_1^2/t$, so $|N_{\mathrm{bad}}| \leqslant C(\sqrt{\delta}/\gamma)|N|$;
- with $S = G$ and (2): $\frac{1}{4}s_1 t \cdot |(N \setminus N_{\mathrm{bad}}) \setminus A_c| \leqslant \int_G |R(e'') \setminus A_c| \, de'' \leqslant C\sqrt{\delta}\, \gamma t^2 \cdot s_1^2/t$.

Together,

$$|N \setminus A_c| \leqslant \varepsilon |N|, \qquad \varepsilon = \varepsilon(\delta, \gamma) = C(1 + \gamma^{-1})\sqrt{\delta}. \tag{3}$$

*Step 5: proof of* (i). Fix $\gamma$ and let $\delta \to 0$ and $\rho \to 0$. Then $|y_1 - y^*| < 2\rho$, $s_1 \to 0$, and $N$ lies in the $(\gamma + 3\rho)$-neighbourhood of $z_1$, while $A_c$ has density $\to 1$ in $N$. So $|A_c \cap B(z_1, 2\gamma)| > 0$. Since $\gamma$ was arbitrary, $z_1 \in H_c$.

*Step 6: proof of* (ii). Let $z = y_1 + \mathbf{e}(\psi_0)$ and $y_2' = y_2 + (y_1 - y^*)$, so $|z - y_2'| = 1$. Since $y_2 \notin \{y^*, 2z_1 - y^*\}$, the angle $\theta = \angle(y_1, z, y_2')$ satisfies $\sin\theta \geqslant \sigma > 0$ uniformly for $y_1$ near $y^*$. Put $L = c_0\sigma\gamma$ with $c_0$ small, and $L \leqslant L_0/2$. For $y \in B(y_2', L)$ the circle $\omega(y)$ crosses $\omega(y_1)$ inside the angular window of $N$ at an angle $\geqslant \sigma/2$, so $\mathcal{H}^1(\omega(y) \cap N) \geqslant s_1/2$. By (W2) with $T = B(y_2', L)$ and $S = N$, for a.e. $y \in T \cap A_c$ the arc $\omega(y) \cap N$ lies in $N \setminus A_c$ up to a null set. Lemma 2.2 and (3) give

$$|B(y_2', L) \cap A_c| \cdot \tfrac{s_1}{2} \leqslant |N \setminus A_c| \cdot 4L \leqslant C\varepsilon\gamma s_1 L.$$

For small $\rho$ we have $B(y_2, L/2) \subset B(y_2', L)$. So the density of $A_c$ in $B(y_2, L/2)$ is at most $C\varepsilon\gamma/L = C\varepsilon/(c_0\sigma)$. Now $L$ does not depend on $\delta$, and $\varepsilon \to 0$ as $\delta \to 0$. Hence $|A_c \cap B(y_2, L/2)| = 0$. □

**Remark 5.2.** The lemma is sharp in the following sense. In a band around a unit circle, colour nine sectors of 40° cyclically $a, b, c$, and replace one $a|b$ boundary by a thin wedge in which $a$ and $b$ are mixed along a fat Cantor-type interface. The colouring stays weak, (Fat) holds in the wedge, its partners lie in $c$-sectors, and its non-degenerate second partners lie on an $a|b$ boundary, where $c$ is null. In the full plane, (i) and (ii) contradict each other (use an equilateral triangle on $y^* z_1$), so a weak 3-colouring of a disc of radius $> 2$ has no fat boundaries near its centre.

5.3. **Empty interior.** From now on $(\Gamma_u, \Gamma_v)$ is a $\kappa$-near bicycle with $\kappa \leqslant \frac{1}{8}$, and $\Omega_0 \subset \Omega_1 \subseteq \Omega$ are as in Section 3. Replacing $\Omega$ by $\Omega_1$, we assume $\Omega = \Omega_1$.

**Theorem 5.3.** $\Sigma_3 \cap \Omega_0$ *has empty interior.*

*Proof.* Suppose $U \subset \Sigma_3 \cap \Omega_0$ is open and non-empty, and let $y^* \in U$. For every ordered pair $(a, b)$, $A_a \cap H_b \cap B(y^*, \varrho) \supseteq A_a \cap B(y^*, \varrho) \cap U$ has positive measure, so (Fat)$(a, b)$ holds at $y^*$. By Corollary 3.7 there are $z_1, y_2$ as in Lemma 5.1: $z_1 \in \Gamma_u \cup \Gamma_v$ has a small arc of $\omega(y^*)$ around it inside $\Omega_1$, and $y_2 \in \Omega_1$ has a disc around it inside $\Omega_1$. Applying Lemma 5.1(ii) to the three pairs $(a, b)$, each colour is null near $y_2$. This contradicts $H_1 \cup H_2 \cup H_3 = \Omega$. □

## 6. $F$-regions, cones, and the dimension of $\Sigma_3$

**Definition 6.1.** An *$F$-region for* $\{a, b\}$ is a non-empty open set $V \subset \Omega$ with $V \subset H_a \cap H_b$ and $V \cap H_c = \emptyset$. Its *partner set* is $N(V) = \{x \in \Omega : \omega(x) \cap V \neq \emptyset\}$, an open set.

**Lemma 6.2.** *Let $V$ be an $F$-region for $\{a,b\}$.*

(i) *$A_a \cap H_b \cap N(V)$ and $A_b \cap H_a \cap N(V)$ are null.*
(ii) *No point of $N(V)$ has $K = \{a\}$ or $K = \{b\}$.*
(iii) *$A_a \cap N(V) \subset H_c$ and $A_b \cap N(V) \subset H_c$ up to null sets.*
(iv) *(Fat)$(a,c)$ holds at a.e. point of $A_a \cap N(V)$, and (Fat)$(b,c)$ at a.e. point of $A_b \cap N(V)$.*

*Proof.* (i) Let $y^* \in N(V)$ and $z_1 \in \omega(y^*) \cap V$. An arc of $\omega(y^*)$ around $z_1$ lies in $V \subset \Omega$. If (Fat)$(a,b)$ held at $y^*$, Lemma 5.1(i) would give $z_1 \in H_c$, which is false. So each point of $N(V)$ has a neighbourhood on which $A_a \cap H_b$ is null; Lindelöf's theorem gives (i).

(ii) If $K(y) = \{a\}$, then by (W4) $a \notin K(z)$ for every partner $z$ of $y$. But $y$ has a partner in $V \subset H_a$.

(iii) The open set $O = N(V) \setminus (H_b \cup H_c)$ satisfies $f = a$ a.e. on $O$, so $K = \{a\}$ on $O$, and $O = \emptyset$ by (ii). So $A_a \cap N(V) \subset H_b \cup H_c$, and (i) removes $H_b$ up to a null set.

(iv) A.e. point of $A_a \cap H_c$ is a density point of this set. □

6.1. **The cone lemma.** Fix an $F$-region $V$ for $\{a,b\}$, a point $x_0 \in N(V)$, and $z_0 \in \omega(x_0) \cap V$ with $B(z_0, 4\beta) \subset V$. Put $\nu = z_0 - x_0$, a unit vector. Choose $0 < \alpha < \beta$, then $\ell_0 < \beta/4$, then $\varepsilon < \ell_0/4$ such that $B(x_0, 4\ell_0) \subset \Omega$. Let

$$C^\circ = \{s\mathbf{e} :\ 0 < s < \ell_0,\ |\mathbf{e}| = 1,\ |\mathbf{e} \cdot \nu| < \sin\alpha\},$$

an open double cone around the line $\nu^\perp$; note $C^\circ = -C^\circ$.

**Lemma 6.3.** *Let $y \in B(x_0, \varepsilon)$ and $p \in y + C^\circ$. Then there is $z_1 \in V$ with $|z_1 - y| = |z_1 - p| = 1$, $p \notin \{y, 2z_1 - y\}$, and $B(z_1, \beta) \subset V$. In particular $B(x_0, \varepsilon) \subset N(V)$.*

*Proof.* Put $s = |p - y| < \ell_0$, and $z_1 = \frac{y+p}{2} + \sqrt{1 - s^2/4}\, n$, where $n \perp p - y$ is the unit vector with $n \cdot \nu > 0$. The angle between $n$ and $\nu$ is $< \alpha$, so

$$|z_1 - z_0| \leqslant |z_1 - (y + \nu)| + |y - x_0| \leqslant \tfrac{s}{2} + \tfrac{s^2}{8} + \alpha + \varepsilon < 3\beta.$$

Hence $B(z_1, \beta) \subset B(z_0, 4\beta) \subset V$. Also $p \neq y$, and $|2z_1 - y - y| = 2 \neq s$. □

**Lemma 6.4** (Cone exclusion)**.** *If $x \in H_a \cap B(x_0, \varepsilon/2)$ and $q \in x + C^\circ$, then $q \notin H_b$. Symmetrically with $a$ and $b$ exchanged.*

*Proof.* Since $C^\circ$ is open, there is $r < \varepsilon/2$ with $q \in y + C^\circ$ for all $y \in B(x, r)$; note $B(x, r) \subset B(x_0, \varepsilon) \subset N(V)$. Since $x \in H_a$, $|A_a \cap B(x, r)| > 0$, and by Lemma 6.2(iv) a.e. $y \in A_a \cap B(x, r)$ satisfies (Fat)$(a,c)$. For such $y$, take $z_1$ from Lemma 6.3 (with $p = q$). Lemma 5.1(ii), applied to the fat pair $(a,c)$ with third colour $b$, $y^* = y$ and $y_2 = q$, shows that $b$ is null near $q$. □

**Lemma 6.5.** *Let $Y \subset B(x_0, \varepsilon/4)$ be a non-degenerate continuum such that every chord direction $\mathbf{e}$ of $Y$ satisfies $|\mathbf{e} \cdot \nu| \geqslant \sin\alpha$. In coordinates $(\xi, \eta)$ along $(\nu^\perp, \nu)$:*

(a) *$Y = \{(g(\eta), \eta) : \eta \in J\}$ with $J$ a non-degenerate compact interval and $\operatorname{Lip} g \leqslant 1/\sin\alpha$; in particular $|Y| = 0$;*
(b) *if $\eta_1$ is the midpoint of $J$, $x_1 = (g(\eta_1), \eta_1)$ and $\rho = \min(|J|/4,\ \ell_0 \sin\alpha/4,\ \varepsilon/4)$, then every $p \in B(x_1, \rho) \setminus Y$ lies in $q + C^\circ$ for some $q \in Y$.*

*Proof.* (a) Along chords $|\Delta\eta| \geqslant \sin\alpha\, |\Delta|$, so $\eta$ is injective on $Y$ and $\xi$ is a $(1/\sin\alpha)$-Lipschitz function of $\eta$. The image $J = \eta(Y)$ is connected and not a point. (b) Let $p = (\xi, \eta) \in B(x_1, \rho) \setminus Y$. Then $\eta \in J$; put $q = (g(\eta), \eta) \in Y$. The vector $p - q = (\xi - g(\eta), 0)$ is non-zero, parallel to $\nu^\perp$, and of length $\leqslant \rho + \rho/\sin\alpha < \ell_0$. □

**Theorem 6.6.** *Let $V$ be an $F$-region for $\{a,b\}$ and $x_0 \in N(V)$, with $\varepsilon$ as above. Then $H_a \cap H_b \cap B(x_0, \varepsilon/4)$ contains no non-degenerate continuum.*

*Proof.* Let $Y \subset H_a \cap H_b \cap B(x_0, \varepsilon/4)$ be a non-degenerate continuum. By Lemma 6.4, $Y$ has no chord with direction in $C^\circ$, so Lemma 6.5 applies. Let $x \in H_a \cap B(x_1, \rho) \setminus Y$. Lemma 6.5(b) gives $q \in Y \subset H_b$ with $x - q \in C^\circ$, so $q \in x + C^\circ$, and $x \in B(x_0, \varepsilon/2)$. This contradicts Lemma 6.4. Hence $H_a \cap B(x_1, \rho) \subset Y$ is null, so $x_1 \notin H_a$. But $x_1 \in Y \subset H_a$. □

Theorem 6.6 uses neither the third colour nor $\Sigma_3$. Isolated points of $H_a \cap H_b \cap N(V)$ are not excluded, and indeed a two-sector configuration around a single point satisfies all cone constraints.

6.2. $\Sigma_3$ **is zero-dimensional.** For a set $\Gamma$ put $\Delta(\Gamma) = \{z : \min_\Gamma |z - \cdot| < 1 < \max_\Gamma |z - \cdot|\}$, an open set when $\Gamma$ is compact. If $\Gamma$ is connected, every $z \in \Delta(\Gamma)$ is a partner of a point of $\Gamma$, by the intermediate value theorem.

**Lemma 6.7.** *If $\Gamma \subset \Omega_0$ contains two points $p \neq q$, then $\Delta(\Gamma) \cap \Omega_0 \neq \emptyset$.*

*Proof.* By Lemma 3.6, $\omega(p)$ meets $\Gamma_u \cup \Gamma_v \subset \Omega_0$, so an arc $A \subset \omega(p) \cap \Omega_0$ exists. It meets $\omega(q)$ in at most two points; pick $z \in A$ with $|z - q| \neq 1$. Moving $z$ slightly along the radius from $p$ (inward if $|z - q| > 1$, outward otherwise) gives a point of $\Omega_0 \cap \Delta(\{p, q\}) \subset \Delta(\Gamma)$. □

**Theorem 6.8.** *$\Sigma_3 \cap \Omega_0$ contains no non-degenerate continuum. Hence $\Sigma_3 \cap K$ is 0-dimensional for every compact $K \subset \Omega_0$.*

*Proof.* Let $\Gamma \subset \Sigma_3 \cap \Omega_0$ be a non-degenerate continuum.

*An $F$-region.* The set $W = \Delta(\Gamma) \cap \Omega_0$ is open and non-empty (Lemma 6.7). Every point of $W$ is a partner of a point of $\Gamma \subset \Sigma_3$, so it has $|K| \geqslant 2$ by (W4). Thus $W$ is covered by the three relatively closed sets $H_a \cap H_b$, and by Baire's theorem one of them has non-empty interior $V_0 \subset W$. By Theorem 5.3, $V = V_0 \setminus H_c$ is non-empty; it is an $F$-region for $\{a, b\}$.

*A small continuum near $V$.* Pick $z \in V$; it is a partner of some $x_0 \in \Gamma$, so $x_0 \in N(V)$. Let $\varepsilon$ be as in the cone lemma, take $0 < r < \min(\varepsilon/4, \operatorname{diam}\Gamma/2)$, and let $Y$ be the component of $\Gamma \cap \overline{B(x_0, r)}$ containing $x_0$. By the boundary bumping theorem $Y$ meets $\partial B(x_0, r)$, so $Y$ is non-degenerate. Also $Y \subset \Sigma_3 \subset H_a \cap H_b$ and $Y \subset B(x_0, \varepsilon/4)$ (after shrinking $r$). This contradicts Theorem 6.6.

*Dimension.* A compact metric space all of whose components are points is 0-dimensional [4]. □

## 7. Removability of $\Sigma_3$

Keep the setting of Theorem 6.8, and put $M_0 = M \cap (\Omega_0 \times \Omega_0)$ and $\Omega_0' = \Omega_0 \setminus \Sigma_3$.

**Lemma 7.1** (Trivial local monodromy)**.** *Every $x \in \Sigma_3 \cap \Omega_0$ has a radius $\rho_x > 0$ with $B(x, \rho_x) \subset \Omega_0$ such that every loop in $B(x, \rho_x) \setminus \Sigma_3$ has trivial monodromy.*

*Proof.* By Lemma 3.6 there is an arc of $\omega(x)$ inside $\Omega_0$. By Theorem 6.8 it is not contained in $\Sigma_3$, so it contains $z_1 \notin \Sigma_3$; choose $\rho$ with $B(z_1, \rho) \subset \Omega_0'$ and put $\rho_x = \rho/2$ (smaller if needed). Let $e = z_1 - x$ and let $\lambda$ be a loop in $B(x, \rho_x) \setminus \Sigma_3$. Monodromy is invariant under free homotopy, so we may move the base point so that $\lambda(0)$ and $\lambda(0) + e$ are good points (an a.e. condition). Then $(\lambda, \lambda + e)$ is a loop in $M'$, and Proposition 4.3 gives $s(\lambda) = s(\lambda + e)$. The loop $\lambda + e$ lies in the disc $B(z_1, \rho) \subset \Omega'$, so $s(\lambda + e) = 0$. □

**Lemma 7.2** (Extension)**.** *There is a function $\tilde{s}$, defined on paths in $\Omega_0$ whose endpoints are good points, such that*

- *$\tilde{s}$ is invariant under homotopy rel endpoints in $\Omega_0$ and additive under concatenation;*
- *$\tilde{s}(\alpha) \equiv f(\alpha(1)) - f(\alpha(0)) \pmod 3$;*
- *$\tilde{s}(\alpha) = s(\alpha)$ when $\alpha \subset \Omega_0'$.*

*Proof.* By Theorem 6.8 and [4, Thm. IV.4], $\Sigma_3$ separates no connected open subset of $\Omega_0$: for every disc, or intersection of two discs, $D \subset \Omega_0$, the set $D \setminus \Sigma_3$ is connected, open, and hence path-connected.

Let $\mathcal{U}$ be the cover of $\Omega_0$ by the discs $B(x, \rho_x)$ ($x \in \Sigma_3 \cap \Omega_0$) and by discs contained in $\Omega_0'$. By Lemma 7.1, every loop in $D \setminus \Sigma_3$, $D \in \mathcal{U}$, has trivial monodromy.

*Pushing off.* Given a path $\alpha$ in $\Omega_0$ with good endpoints, subdivide it so that consecutive pieces lie in members $D_i \in \mathcal{U}$. Replace each subdivision point by a nearby good point of $D_{i-1} \cap D_i$ and join consecutive points inside $D_i \setminus \Sigma_3$. Straight-line homotopies in the convex sets $D_i$ show that the new path $\alpha' \subset \Omega_0'$ is homotopic to $\alpha$ rel endpoints in $\Omega_0$. Set $\tilde{s}(\alpha) = s(\alpha')$.

*Well defined.* Let $\alpha', \alpha'' \subset \Omega_0'$ be homotopic rel endpoints in $\Omega_0$, by $h\colon [0,1]^2 \to \Omega_0$. Subdivide the square into small squares, each mapped into some $D \in \mathcal{U}$ (Lebesgue number). Map each interior grid vertex to a nearby good point, and each interior grid edge to a path in $(D \cap D') \setminus \Sigma_3$, where $D, D'$ are the members assigned to the adjacent squares. Then the loop $\alpha' \cdot \overline{\alpha''}$ is a product of conjugates of loops each contained in a single $D \setminus \Sigma_3$. These have trivial monodromy, so $s(\alpha') = s(\alpha'')$.

The listed properties follow from those of $s$. □

**Lemma 7.3** (Pushing pair paths)**.** *Let $Z = \pi_1^{-1}(\Sigma_3) \cup \pi_2^{-1}(\Sigma_3) \subset M_0$. Every path in $M_0$ whose endpoints lie in $M_0 \setminus Z$ is homotopic rel endpoints in $M_0$ to a path in $M_0 \setminus Z$, by a homotopy whose projections stay in $\Omega_0$.*

*Proof.* Via $(x, y) \mapsto (x, \arg(y - x))$, $M_0$ is a 3-manifold, and over a compact set $Z$ is a union of two sets homeomorphic to closed subsets of $(\Sigma_3 \cap K) \times S^1$. So $\dim Z \leqslant 1 = 3 - 2$, and by [4, Thm. IV.4] $Z$ separates no connected open subset of $M_0$. Now argue exactly as in the pushing-off step of Lemma 7.2, using small convex coordinate balls in $M_0$. □

**Proposition 7.4** (Index estimate across $\Sigma_3$)**.** *Let $\gamma$ be a path in $M_0$ whose endpoints are pairs of good points. Then $|\tilde{s}(\pi_1\gamma) - \tilde{s}(\pi_2\gamma)| \leqslant 1$.*

*Proof.* By Lemma 7.3, $\gamma$ is homotopic rel endpoints in $M_0$ to a path $\gamma'$ in $M' \cap M_0$. Proposition 4.3 applies to $\gamma'$. Since $\pi_i\gamma'$ is homotopic to $\pi_i\gamma$ rel endpoints in $\Omega_0$, Lemma 7.2 gives $\tilde{s}(\pi_i\gamma) = s(\pi_i\gamma')$. □

## 8. Proof of the Main Theorem

**Theorem 8.1.** *For every $d \in (1,2)$ there is $\kappa_0(d) > 0$ such that, if $(\Gamma_u, \Gamma_v)$ is a $\kappa$-near bicycle with $\kappa \leqslant \kappa_0$, $\delta > 0$, and $\Omega \supseteq N_\delta(\Gamma_u \cup \Gamma_v)$ is open, then $\Omega$ admits no weak measurable 3-colouring.*

*Proof.* Let $\kappa_0 = \kappa_0(d)$ be as in Proposition 3.4, and suppose $f$ is a weak 3-colouring of $\Omega$. We may assume $\delta \leqslant \frac{1}{8}$ and $\Omega = \Omega_1$, and we use $\Omega_0$, $\Sigma_3$, $\tilde{s}$ as in Sections 4–7.

*Translation.* For $|w| < \delta/4$ the translated pair $(\Gamma_u + w, \Gamma_v + w)$ is again a $\kappa$-near bicycle, and $\Gamma_u + w, \Gamma_v + w \subset \Omega_0$. Let $p, q_\pm$ be the points of Proposition 3.4 for $(\Gamma_u, \Gamma_v)$; the points for the translated pair are $p + w, q_\pm + w$, and all paths and homotopies of Proposition 3.4 translate. The set of $w$ for which $p + w, q_\pm + w$ lie outside $\Sigma_3$ is open and dense in $B(0, \delta/4)$ (Theorem 5.3; $\Sigma_3$ is closed). The set of $w$ for which they are Lebesgue points has full measure. Pick $w$ in both, and rename the translated objects $\Gamma_u, \Gamma_v, p, q_\pm, \eta_1, \eta_2, \Lambda, (z, z')$. So $p, q_\pm$ are good points in $\Omega_0$, and $\Gamma_u \cup \Gamma_v \subset \Omega_0$.

*Lower bound.* Since $|p - q_\pm| = 1$ and these are Lebesgue points, (W3) gives $f(q_-) \neq f(p) \neq f(q_+)$. Hence

$$a := \tilde{s}(\eta_1) \not\equiv 0, \qquad b := \tilde{s}(\eta_2) \not\equiv 0 \pmod 3.$$

The path $(z, z')$ of Proposition 3.4(ii) is a path in $M_0$ from $(q_-, p)$ to $(p, q_+)$ with $z = \eta_1$ and $z' \simeq \eta_2$ rel endpoints in $\Gamma_v \subset \Omega_0$. Proposition 7.4 and Lemma 7.2 give $|a - b| \leqslant 1$. Two integers

prime to 3 at distance $\leqslant 1$ have the same sign (neither is 0), and each has absolute value $\geqslant 1$. Therefore

$$|\tilde{s}(\eta)| = |a + b| \geqslant 2.$$

*Upper bound.* The path $\Lambda$ of Proposition 3.4(i) is a path in $M_0$ from $(p, q_-)$ to $(p, q_+)$. Its first projection is a loop at $p$, null-homotopic in $\Gamma_u \subset \Omega_0$, so $\tilde{s}(\pi_1\Lambda) = 0$ by Lemma 7.2. Its second projection is homotopic to $\eta$ rel endpoints in $\Gamma_v \subset \Omega_0$, so $\tilde{s}(\pi_2\Lambda) = \tilde{s}(\eta)$. Proposition 7.4 gives

$$|\tilde{s}(\eta)| = |\tilde{s}(\pi_2\Lambda) - \tilde{s}(\pi_1\Lambda)| \leqslant 1.$$

This contradicts the lower bound. □

*Proof of the Corollary.* Two unit circles $T(u), T(v)$ with $1 < |u - v| < 2$ form a 0-near bicycle. □

**Remark 8.2** (Regular colourings)**.** If the colour classes have locally finite perimeter, the proof simplifies drastically. By the coarea formula and BV slicing, for a.e. translation $w$ the restriction of $f$ to $\Gamma_u + w$ and $\Gamma_v + w$ has finitely many jumps, and (W1) holds on the relevant circles. The classical step counts of [9, 6] are then defined directly, and the argument above runs without Sections 5–7. The whole difficulty of the general case is the set $\Sigma_3$, which may have positive measure a priori and is shown here to be 0-dimensional inside the band.

**Concluding remarks.**

(1) The Main Theorem is the measurable form of step (SV3). As explained in the introduction, it does not by itself decide the Hadwiger–Nelson problem: a measurable analogue of (SV1), (SV2) and, above all, the supply of trichromatic points and step (SV4) are still missing.

(2) The shell lemma and the cone lemma are local statements about arbitrary weak 3-colourings and may be of independent interest. For instance, they imply that weak 3-colourings of a disc of radius $> 2$ have no fat interfaces near the centre.

(3) The constant $\kappa_0(d)$ degenerates as $d \to 1$ or $d \to 2$. For $d \to 2$ this reflects the tangency of the exact circles; we do not know whether the theorem extends to configurations with tangencies.

## References


[1] A. D. N. J. de Grey, *The chromatic number of the plane is at least 5*, Geombinatorics **28** (2018), 18–31.

[2] G. Exoo and D. Ismailescu, *The chromatic number of the plane is at least 5: a new proof*, Discrete Comput. Geom. **64** (2020), 216–226.

[3] K. J. Falconer, *The realization of distances in measurable subsets covering* $\mathbb{R}^n$, J. Combin. Theory Ser. A **31** (1981), 184–189.

[4] W. Hurewicz and H. Wallman, *Dimension Theory*, Princeton University Press, 1941.

[5] OpenAI, *The Euclidean plane is not five-colorable*, preprint (2026), https://github.com/openai/math.

[6] D. Sokolov and V. Voronov, *On the chromatic number of the plane for map-type colorings*, arXiv:2502.01958 (2025).

[7] S. P. Townsend, *Every* 5*-coloured map in the plane contains a monochrome unit*, J. Combin. Theory Ser. A **30** (1981), 114–115.

[8] S. P. Townsend, *Colouring the plane with no monochrome units*, Geombinatorics **14** (2005), 181–193.

[9] V. Voronov, *On the chromatic number of the plane with an interval of forbidden distances*, arXiv:2304.10163 (2023).

[10] D. R. Woodall, *Distances realized by sets covering the plane*, J. Combin. Theory Ser. A **14** (1973), 187–200.